

AutoML-NestedCV-MATLAB

A MATLAB-based AutoML GUI implementing repeated nested cross-validation with hyperparameter optimization for binary and multi-class classification. Features leakage-free preprocessing, SHAP analysis, ROC/AUC evaluation, and internal/external validation with statistical comparison.

Features

- **Repeated Nested Cross-Validation** with Bayesian hyperparameter optimization
 - **Leakage-free preprocessing** — imputation, normalization, and feature selection applied fold-locally
 - **10 classification algorithms** — white-box and black-box models
 - **SHAP analysis** — marginal contribution-based feature importance
 - **Internal & external validation** — ROC curves, confusion matrices, boxplots with Wilcoxon signed-rank test
 - **Calibration curves** and performance metric comparison across models
 - **GUI interface** — no coding required to run analyses
-

Supported Models

White-Box

Decision Tree (DT)

Logistic Regression (LR)

Discriminant Analysis (DA)

Naive Bayes (NB)

Generalized Additive Model (GAM)

Generalized Linear Model (GLM)

Black-Box

K-Nearest Neighbors (KNN)

Support Vector Machine (SVM)

Neural Network (NN)

Ensemble (ENS)

Requirements

- MATLAB R2021a or later
- Statistics and Machine Learning Toolbox

- (Optional) Deep Learning Toolbox — for Neural Network models
-

Installation

```
bash

git clone https://github.com/YOUR_USERNAME/AutoML-NestedCV-MATLAB.git
```

Then in MATLAB:

```
matlab

cd AutoML-NestedCV-MATLAB
ML_Auto_GUI
```

Usage

1. Launch the GUI by running `ML_Auto_GUI` in MATLAB
 2. Click **Load Dataset** and select your `.csv`, `.xlsx`, or `.mat` file
 - Last column must be the class label
 3. Set analysis parameters:
 - **K-Fold** — number of outer folds (recommended: 10)
 - **Iterations** — number of repeated CV runs (recommended: 10)
 - **Holdout (%)** — external validation set size (recommended: 20)
 4. Select one or more classification models
 5. Click **START ANALYSIS**
 6. After completion, use the dropdown menus to plot:
 - ROC curves (internal & external)
 - Confusion matrices
 - SHAP feature importance
 - Calibration curves
 - Performance metric comparison
-

Output

Each model produces a 2×3 validation figure containing:

	Col1	Col2	Col3
Row 1	Internal ROC	External ROC	AUC Boxplot + Wilcoxon p
Row 2	Internal CM	External CM	ACC Boxplot + Wilcoxon p

Results are also exportable as an Excel file containing mean $\pm$ std for all metrics (ACC, AUC, Sensitivity, Specificity, PPV, NPV, F1, MCC).

File Structure

```
AutoML-NestedCV-MATLAB/
├─ ML_Auto_GUI.m           # Main GUI class
├─ ML_Repeated_Nested_CV.m # Core CV and modeling engine
└─ README.md
```

Citation

If you use this code in your research, please cite:

```
[Your citation will appear here upon publication]
```

License

MIT License — see [LICENSE](#) for details.